Foundations of ADCS

1. Preliminaries of ADCS

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April 16, 2023

Foundations of ADCS

https://github.com/risherlock/ADCS

0. Course introduction
1. **Preliminaries of ADCS**
2. Attitude determination system
3. Onboard models
4. Attitude control system
5. ADCS simulation
6. Discussions and conclusion

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2. General control system architecture
3. Measurement
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5. Coordinate frames in ADCS
6. Attitude parameterization
7. Conclusion

Today's goal

We will establish the language necessary to speak ADCS. In particular, we will answer two important questions:

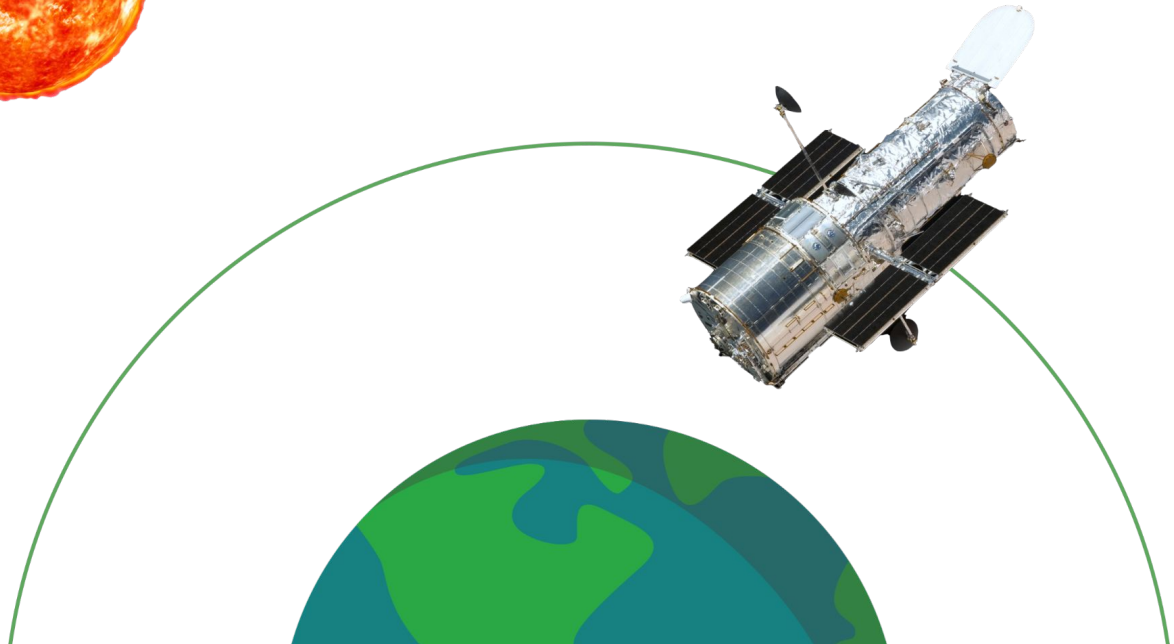
What exactly do we mean by attitude?
How is attitude quantified?

altitude ≠ attitude = orientation

attitude

(noun) the position of the body or way of carrying oneself

Review: Problem statement



Review: Problem statement

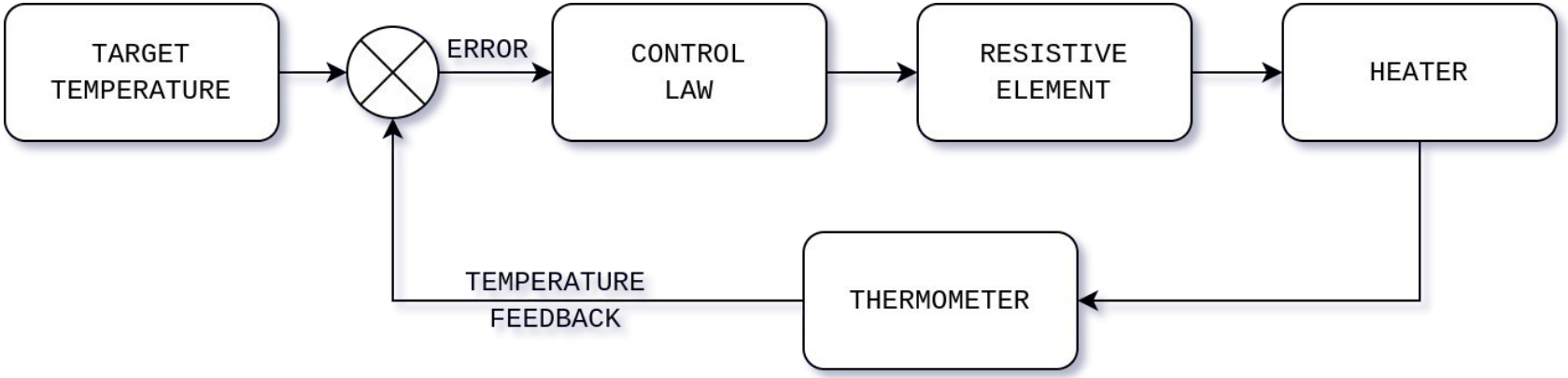
Steps involved in ADCS

1. Attitude measurement - Attitude determination
2. Take action to achieve desired attitude - Attitude control

Let us see how these steps fits into the general picture of feedback control architecture.

General control system architecture

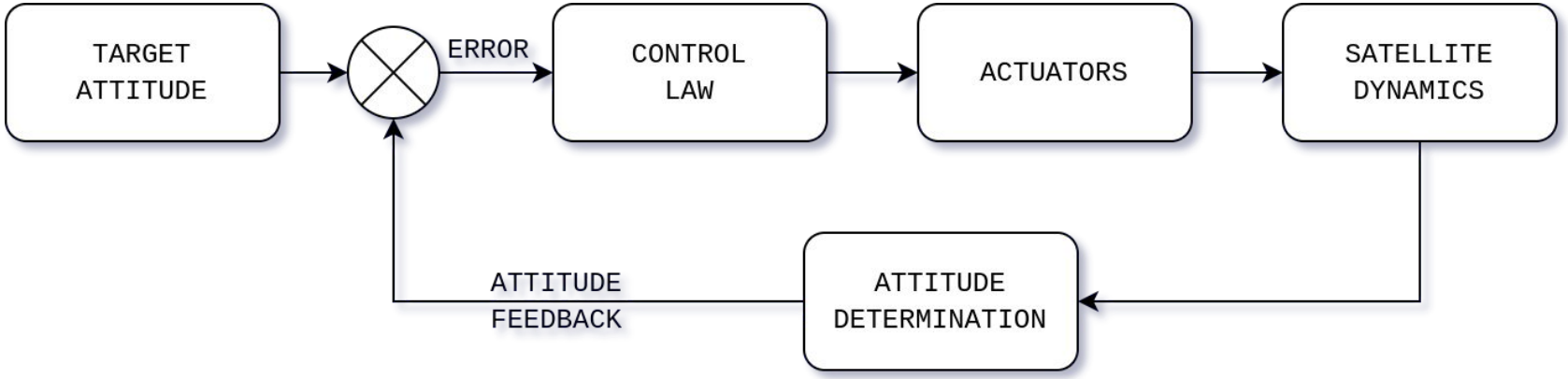
***Temperature* control system**



plant - heater
feedback - temperature
control law - in software
actuator - resistive element

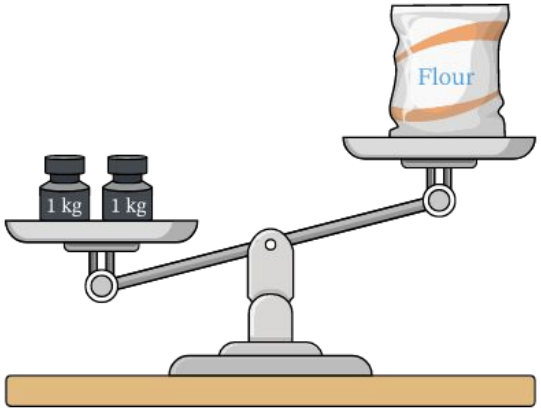
General control system architecture

Attitude control system

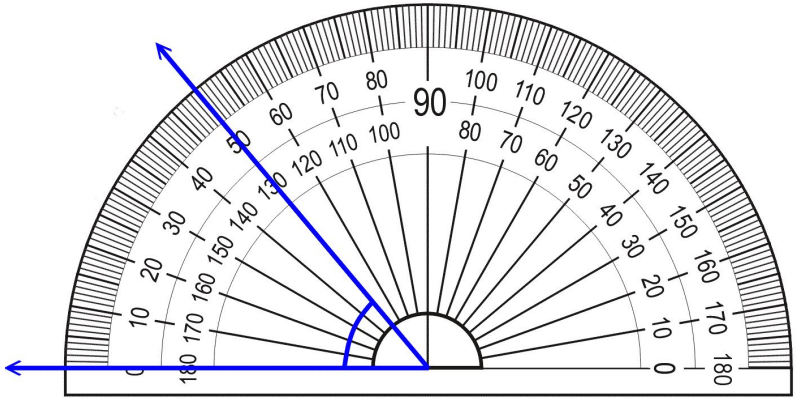


plant - satellite dynamics
feedback - attitude
control law - in software
actuator - reaction wheel, thruster

Measurement



The weight of flour in the packet is less wrt. to 2kg



The upper line is 50° wrt. to the base line

Measurement is a process of determining *how large or small* a physical quantity is as *compared to a basic reference* quantity of the same kind.

- Wikipedia

Measurement

Measurement involves *numerical parameterization* relative to a *reference*

Weight measurement

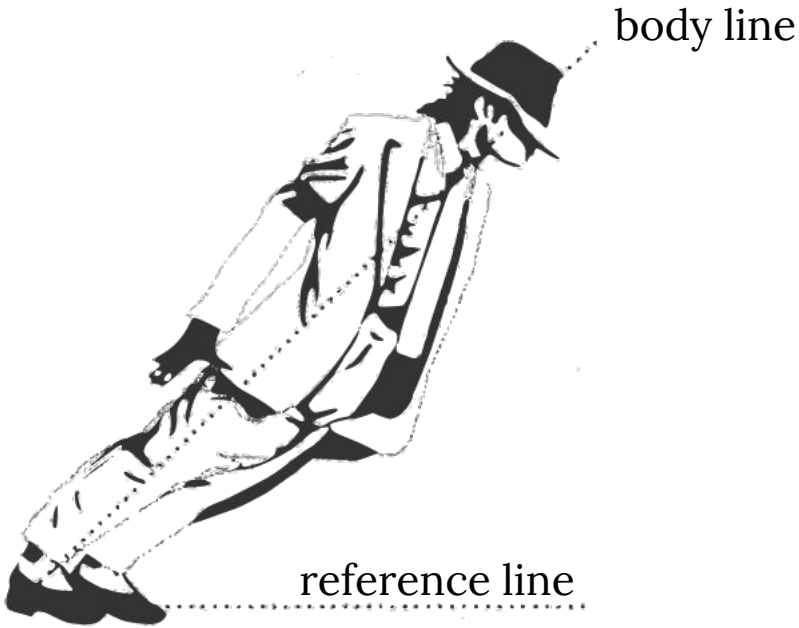
- parameterization - kg
- reference - the mass assigned 1 kg

Angle measurement

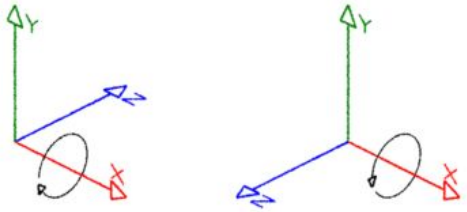
- parameterization - degree
- reference - the base line

Let us establish the reference and parameterization for attitude measurement.

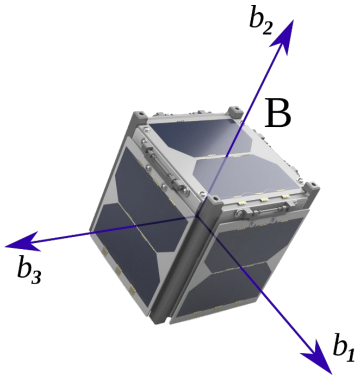
Different references



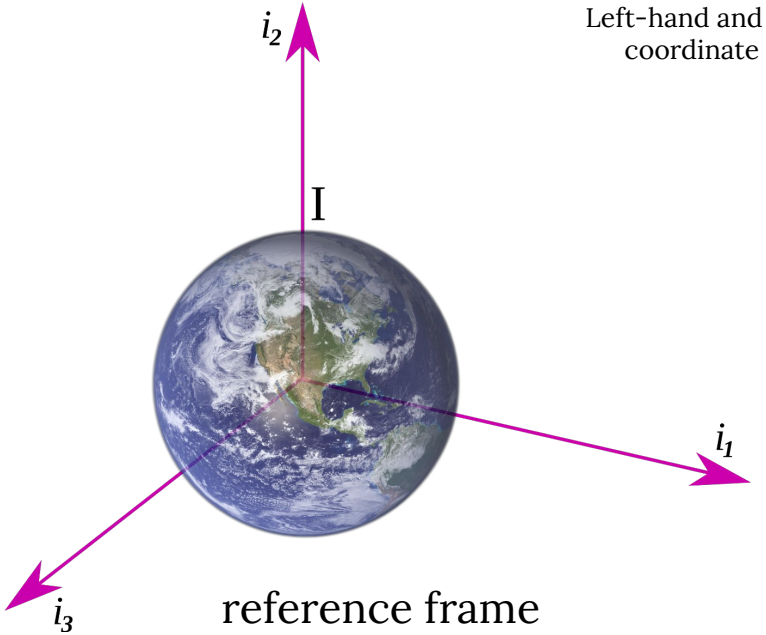
Coordinate frames



Left-hand and right-hand coordinate systems



body frame



reference frame

Coordinate frames in ADCS

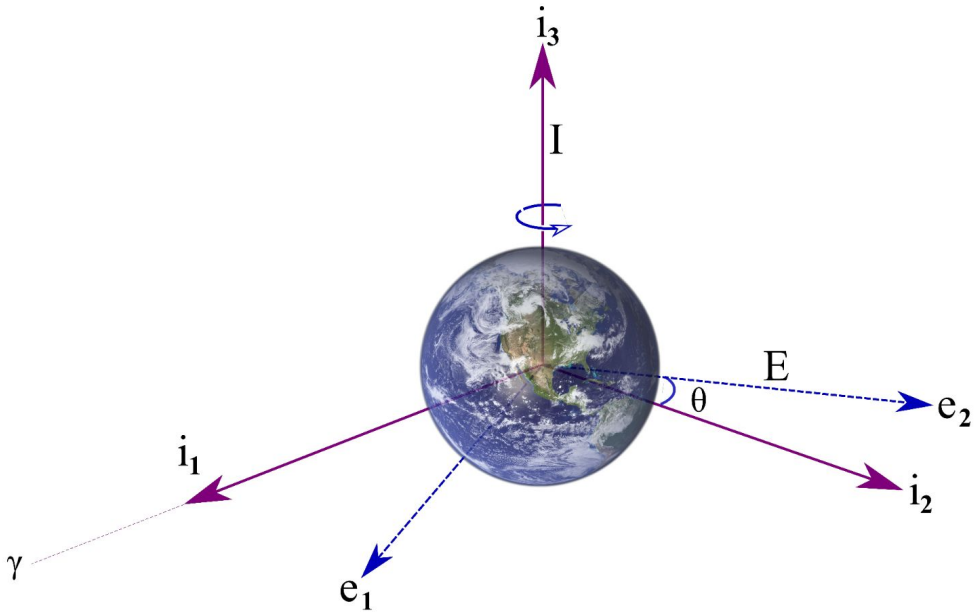
There are different units of weights like ounce, pound, ton, kg etc. They can be used interchangeably with certain mathematical relations existing between them.

Based on the context, some units are more relevant than other and makes the computations easier. For example, in retail shop, kg is more appropriate than ton to measure the weight of sugar.

Similarly, there are various coordinate frames that are used in ADCS.

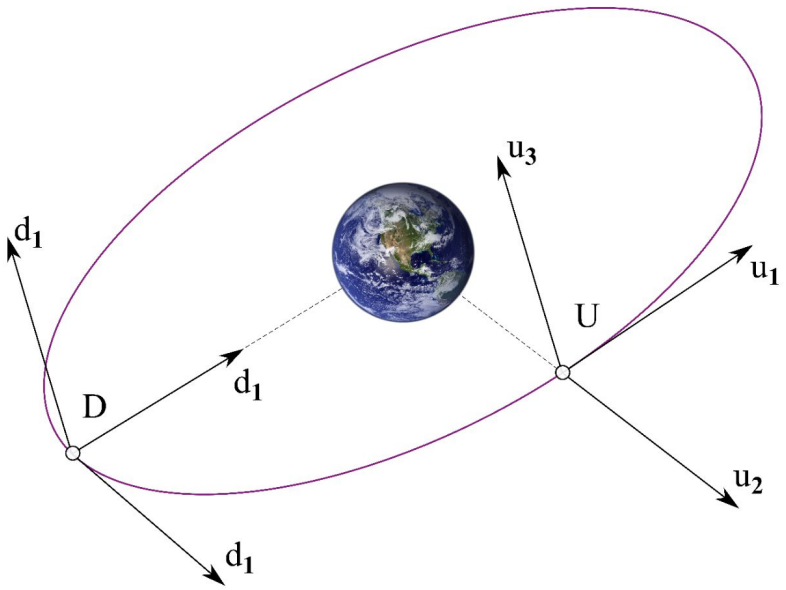
Specific choices depends upon the nature of the problem in consideration. Suitable choice of coordinate frame reduces the computations and makes problem easier to solve.

Frames of references in ADCS



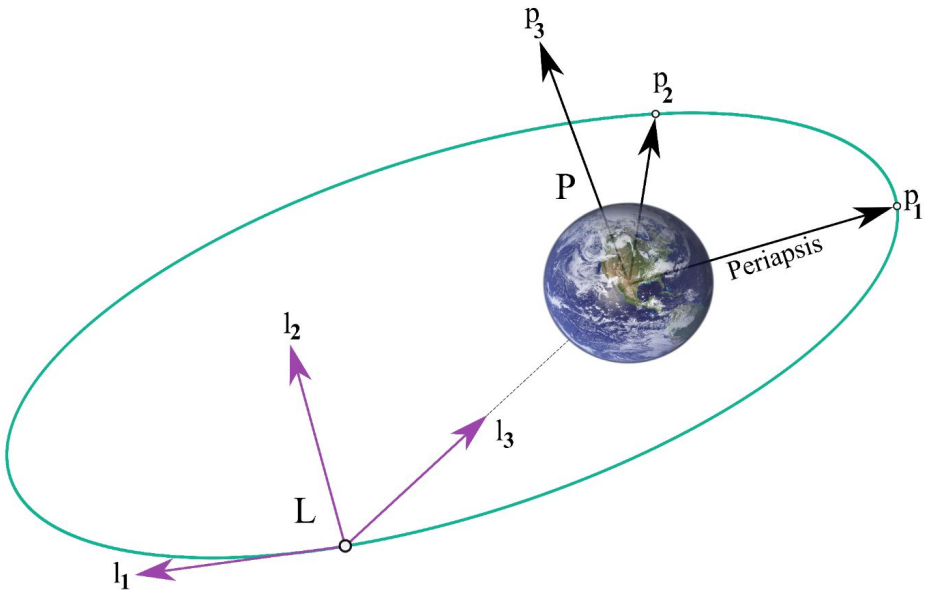
Earth Centered Inertial Frame (ECI) and *Earth Centered Earth Fixed Frame* (ECEF)

Frames of references in ADCS



Local Vertical Local Horizontal (LVLH) frames

Frames of references in ADCS

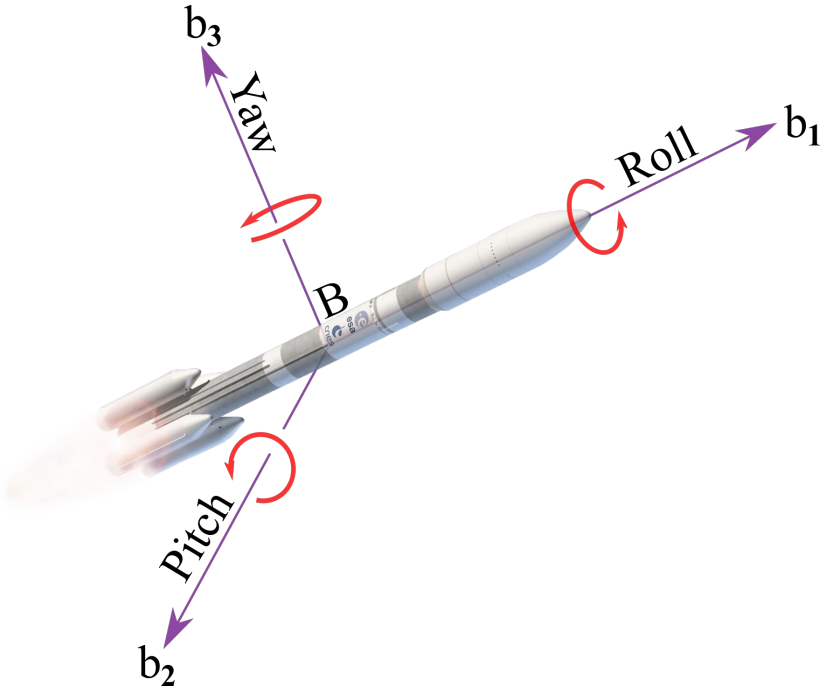


Perifocal frame

Attitude parameterization

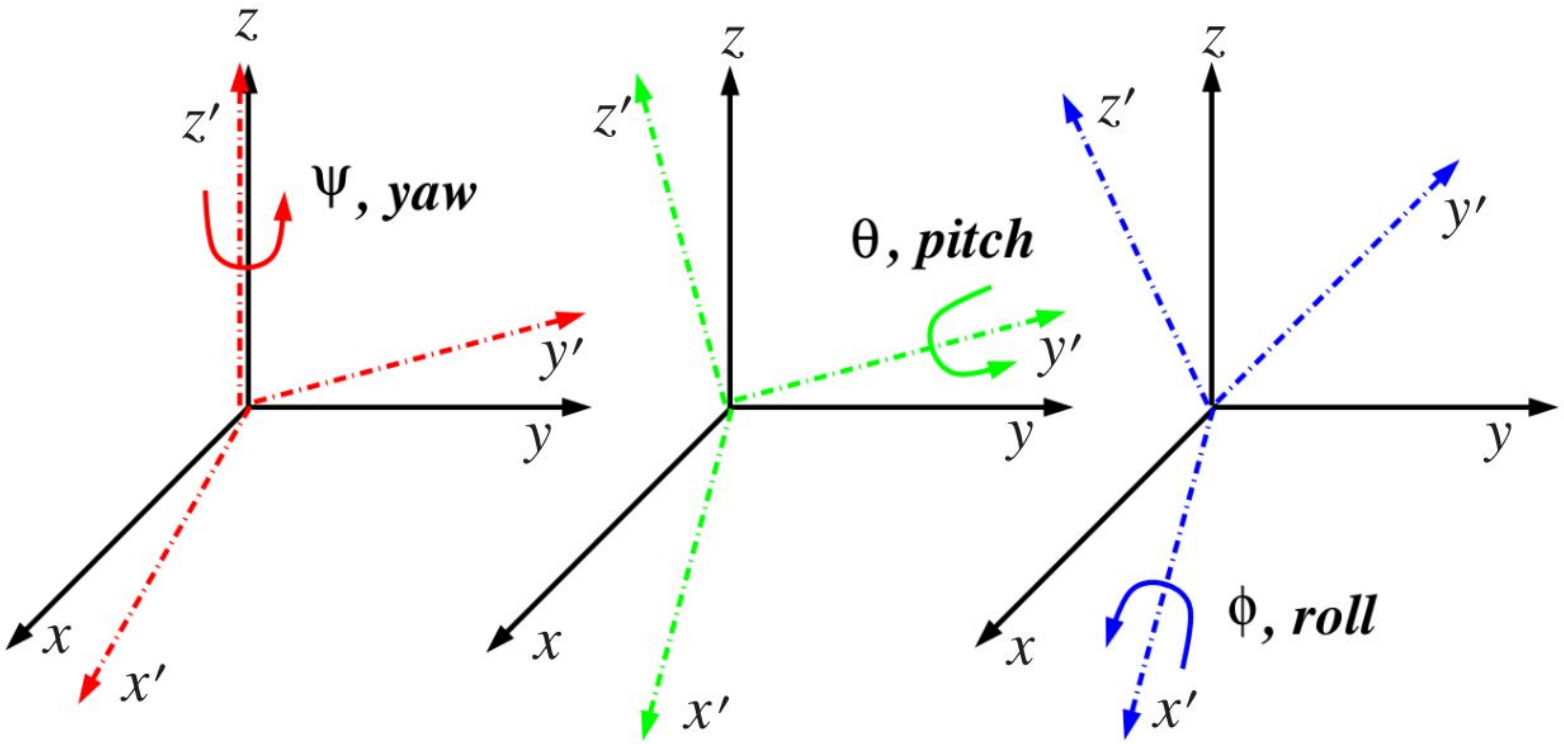
How do we *quantify* or *compare* attitude using numbers?

Euler angles



Arbitrary rotation can be expressed as the result of consecutive rotations about each coordinate axes.

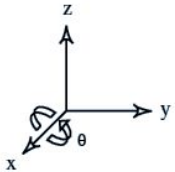
321 Euler sequence



Rotation matrices

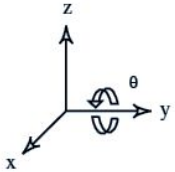
Rotation around the x-Axis

$$\mathbf{P}_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$



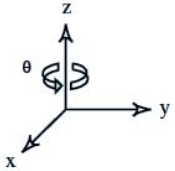
Rotation around the y-Axis

$$\mathbf{P}_y = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$



Rotation around the z-Axis

$$\mathbf{P}_z = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$



$$\boxed{\mathbf{M}^{-1} = \mathbf{M}^{T} \text{ and } |\mathbf{M}| = 1}$$

Matrix multiplication

An important property of rotation matrices is that we can represent consecutive rotations by matrix multiplication of matrices representing each individual rotation.

Example

The rotation matrix corresponding to the consecutive rotations yaw 20°, pitch 10°, and roll 30° (YPR sequence) is

$$\mathbf{R} = \mathbf{P}_x(30°)\, \mathbf{P}_y(10°)\, \mathbf{P}_z(20°).$$

Coordinate transformation

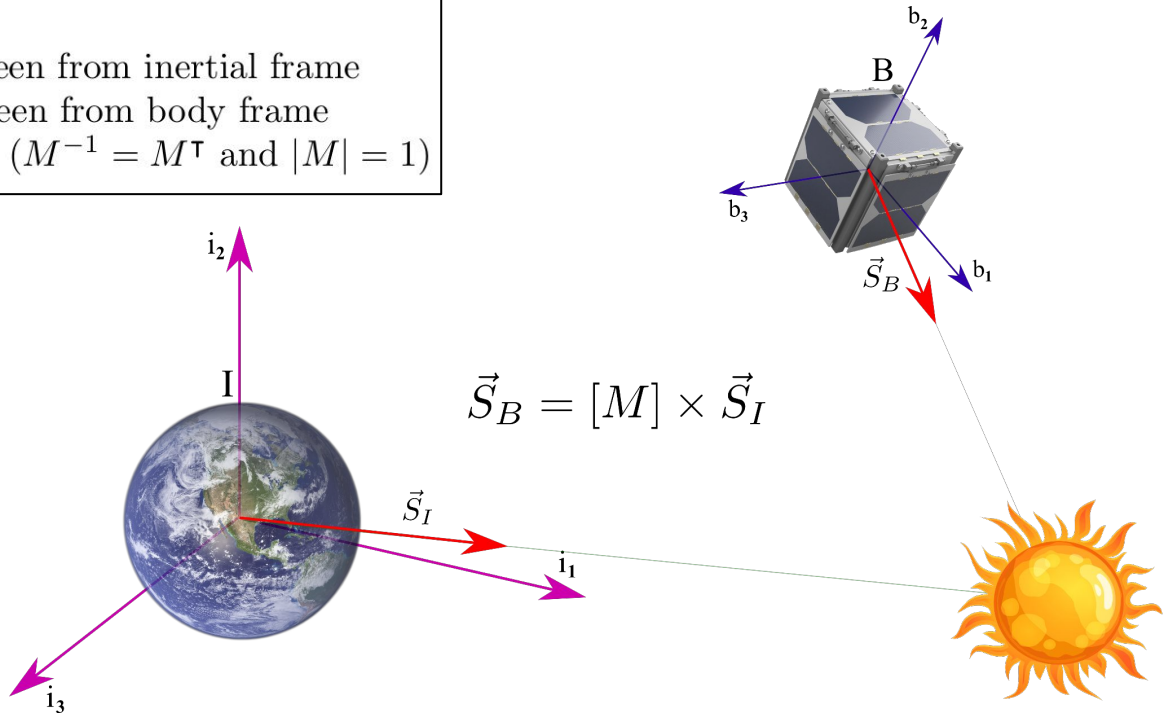
$$\boldsymbol{v}_b = \boldsymbol{M}\boldsymbol{v}_i$$

where,

$\boldsymbol{v}_i \in \mathbb{R}^2$ is a vector as seen from inertial frame

$\boldsymbol{v}_b \in \mathbb{R}^2$ is a vector as seen from body frame

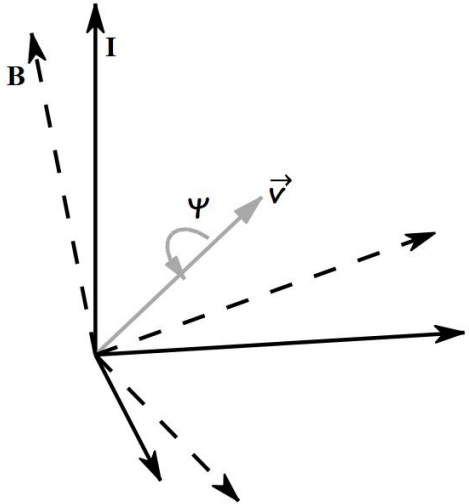
$\boldsymbol{M}$ is a rotation matrix ($M^{-1} = M^{\intercal}$ and $|M| = 1$)



$$\vec{S}_B = [M] \times \vec{S}_I$$

Quaternions: History

Leonhard Euler



An arbitrary rotation **R** in 3D is a rotation around some axis ***v*** by some angle Ψ. ***v*** is a unit vector because its magnitude if is of no consequence.

Quaternions: History

Euler proved that any arbitrary three axis rotation can be expressed as rotation about some arbitrary axis $\boldsymbol{v}$ by an angle $\boldsymbol{\psi}$.

This means that we can parameterize rotation in a new way if we know the three components of the axis $\boldsymbol{v}$ (i.e. v_1, v_2, v_3) and the angles $\boldsymbol{\psi}$. This is known as *angle-axis* representation.

Euler proved the existence of a angle-axis for any arbitrary rotation, but he did not solve it. He could not figure out what that axis and angle would be for given rotation. The problem was solved 64 years later by Rodrigues.

References

Leonard Euler - *Formulae generales pro translatione quacunque corporum rigidorum* (1776)
Translation: *General formulas for the translation of arbitrary rigid bodies*

Quaternions: History (contd..)

Olinde Rodrigues

Let $\boldsymbol{Q}(\psi\boldsymbol{v})$ be the axis-angle representation of the rotation around an axis $\boldsymbol{v} \in \mathbb{R}^3$ by an angle of ψ. Euler's proof allows us to express any arbitrary rotation in angle-axis form.

Let $\boldsymbol{Q}(\psi\boldsymbol{v})$ be the rotation which results after two consecutive rotations $\boldsymbol{Q}(\alpha\boldsymbol{l})$ and $\boldsymbol{Q}(\beta\boldsymbol{m})$. Rodrigues derived the equations for representing the rotation $\boldsymbol{Q}(\psi\boldsymbol{v})$ in terms of the other two.

$$\cos\frac{\psi}{2} = \cos\frac{\alpha}{2}\cos\frac{\beta}{2} - \sin\frac{\alpha}{2}\sin\frac{\beta}{2}$$
$$\sin\frac{\psi}{2}\boldsymbol{v} = \sin\frac{\alpha}{2}\cos\frac{\beta}{2}\boldsymbol{l} + \cos\frac{\alpha}{2}\sin\frac{\beta}{2}\boldsymbol{m} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2}(\boldsymbol{l}\times\boldsymbol{m})$$

where '$\times$' represents the cross product between two vectors.

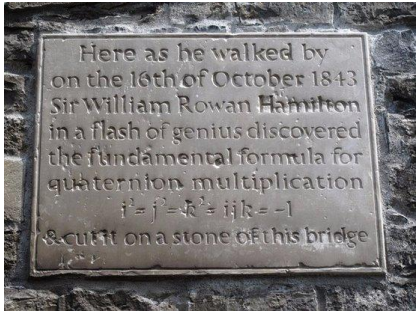
Quaternions: History (contd..)

So far in the history, we saw that Euler proved a theorem which gave a new parameterization of rotation in axis-angle representation. Rodrigues computed the expressions for that axis and angle in the form. It was the discovery of Hamilton which would wrap things up beautifully.

Hamilton was searching for the number system like complex number but with four elements. He discovered such number system, known as *quaternions* by establishing the quaternion multiplication.

The multiplication rule which Hamilton discovered would match the rotation formula derived by Rodrigues if the quaternion were written in the form $[\cos(\boldsymbol{\psi}/2), \sin(\boldsymbol{\psi})\, \boldsymbol{v}]$.

Quaternions: History (contd..)



Rowan Hamilton

A quaternion $\boldsymbol{a} = a_0 + a_1\boldsymbol{i} + a_2\boldsymbol{j} + a_3\boldsymbol{k}$ is a four dimensional number where $\boldsymbol{i}^2 = \boldsymbol{j}^2 = \boldsymbol{k}^2 = \boldsymbol{ijk} = -1$. This relation between $\boldsymbol{i}, \boldsymbol{j}$, and $\boldsymbol{k}$ was discovered by Hamilton which allowed the multiplication of quaterions to be defined leading to the generalization of complex numbers to higher dimension.

There are various ways of representing quaternion.

Algebraic expression: $\boldsymbol{a} = a_0 + a_1\boldsymbol{i} + a_2\boldsymbol{j} + a_3\boldsymbol{k}$

Vector expression: $\boldsymbol{a} = [a_0, \boldsymbol{v}_a]^\intercal$ where, $\boldsymbol{v}_a = [a_1, a_2, a_3]^\intercal$

Quaternions: History (contd..)

$$\cos\frac{\psi}{2} = \cos\frac{\alpha}{2}\cos\frac{\beta}{2} - \sin\frac{\alpha}{2}\sin\frac{\beta}{2}$$
$$\sin\frac{\psi}{2}\boldsymbol{v} = \sin\frac{\alpha}{2}\cos\frac{\beta}{2}\boldsymbol{l} + \cos\frac{\alpha}{2}\sin\frac{\beta}{2}\boldsymbol{m} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2}(\boldsymbol{l}\times\boldsymbol{m})$$

Rowan Hamilton

Let $\boldsymbol{a} = [a_0, \boldsymbol{v}_a]^\intercal$ and $\boldsymbol{b} = [b_0, \boldsymbol{v}_b]^\intercal$ be two quaternions. Hamilton discovered quaternion multiplication in the following form.

$$\boldsymbol{a} \circ \boldsymbol{b} = \begin{bmatrix} a_0 b_0 - \boldsymbol{v}_a \boldsymbol{v}_b \\ a_0 \boldsymbol{v}_b + b_0 \boldsymbol{v}_a + \boldsymbol{v}_a \times \boldsymbol{v}_b \end{bmatrix}$$

If we assume that the resulting quaternion to be $\boldsymbol{c} = [c_0, \boldsymbol{v}_c]^\intercal$, we can write the scalar and vector parts as following.

$$c_0 = a_0 b_0 - \boldsymbol{v}_a \boldsymbol{v}_b$$
$$\boldsymbol{v}_c = a_0 \boldsymbol{v}_b + b_0 \boldsymbol{v}_a + \boldsymbol{v}_a \times \boldsymbol{v}_b$$

Now funny thing about these expression is that if we change the notation of quaternion to be $[\cos\psi/2, \sin[\psi/2]\boldsymbol{v}_c]$ from $[c_0, \boldsymbol{v}_c]^\intercal$, we will see that the quaternion multiplication corresponds to the equations derived by Rodrigues!

This shows that the quaternions discovered by Hamilton represents the axis-angle formalism which was proved and solved by Euler and Rodrigues respectively.

Quaternions and rotation

$$\boldsymbol{q} = \left[\cos\frac{\psi}{2},\ \sin\frac{\psi}{2}\,\boldsymbol{v}^{\mathsf{T}} \right]$$

From Rodrigues' solution and the equation of quaternion multiplication, we can conclude that a quaternion represents rotation of angle Ψ about the axis $\boldsymbol{v}$ when quaternions are expressed as above.

The quaternion multiplication, like multiplications of rotation matrices, represents consecutive rotations.

Note: Only unit quaternions ($|\boldsymbol{q}| = 1$) represent orientation. Quaternions with arbitrary 4 numbers without unit norm constraint does not represent orientation.

Visualizing quaternions

$$\boldsymbol{q} = \left[\cos\frac{\psi}{2},\ \sin\frac{\psi}{2}\,\boldsymbol{v}^{\intercal} \right]$$

Some special cases to be comfortable with how quaternions represents attitude.

1. No rotation about any axis
2. 90° rotation about X-axis
3. 90° rotation about Y-axis
4. 90° rotation about Z-axis

Visualizing quaternions

$$\boldsymbol{q} = \left[\cos\frac{\psi}{2},\ \sin\frac{\psi}{2}\,\boldsymbol{v}^{\intercal} \right]$$

Some special cases to be comfortable with how quaternions represents attitude.

1. No rotation about any axis : [1, 0, 0, 0]
2. 90° rotation about X-axis : $[2^{-0.5}, 2^{-0.5}, 0, 0]$
3. 90° rotation about Y-axis : $[2^{-0.5}, 0, 2^{-0.5}, 0]$
4. 90° rotation about Z-axis : $[2^{-0.5}, 0, 0, 2^{-0.5}]$

Quaternions: Pros

1. Differential equation governing Euler angles becomes infinity when cos θ is zero which creates singularity. Physically, this problem appears as gimbal lock. Quaternion does not have such problem.
2. The equations governing Euler angles consists of transcendental functions which are computationally expensive. Quaternion computations are faster.
3. Quaternion has only one constraint i.e. unit norm. Whereas rotation matrix has nine for it to be an orthogonal matrix.
4. Quaternion multiplication implies two successive rotations.
5. Quaternion rotations are unique. 3 Euler angles might represent different orientation based on the sequence.

$$\begin{pmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \frac{1}{\cos\theta} \begin{bmatrix} 0 & \sin\phi & \cos\phi \\ 0 & \cos\phi\cos\theta & -\sin\phi\cos\theta \\ \cos\theta & \sin\phi\sin\theta & \cos\phi\sin\theta \end{bmatrix}^{\mathcal{B}} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

Fig. Equation governing Euler angles

$$\begin{pmatrix} \dot{\beta}_0 \\ \dot{\beta}_1 \\ \dot{\beta}_2 \\ \dot{\beta}_3 \end{pmatrix} = \frac{1}{2} \begin{bmatrix} \beta_0 & -\beta_1 & -\beta_2 & -\beta_3 \\ \beta_1 & \beta_0 & -\beta_3 & \beta_2 \\ \beta_2 & \beta_3 & \beta_0 & -\beta_1 \\ \beta_3 & -\beta_2 & \beta_1 & \beta_0 \end{bmatrix} \begin{pmatrix} 0 \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

Fig. Equation governing Quaternions

Conclusion

1. Any consideration of attitude requires at least two coordinate frames: a body frame and a reference frame
2. Attitude of body frame is expressed relative to the reference frame.
3. Attitude is can be parameterized by (but not limited to) Euler angles, rotation matrices, and quaternions
4. Out of the three quaternions have more attractive features
5. These attitude parameters are the feedback in attitude control system

In next lecture we will see how attitude parameters are computed (or determined, or estimated)

Thank you!